33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10004

F13-SR-001
415V SAG MILL SWITCHROOM

F13-TR-002
TRANSFORMER ROOM

F13-TR-002
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
MCC EARTH BAR
NEUTRAL BAR

4000/ 1A
2.5VA
5P20

F13-SW-002
SAG MILL 415V MOTOR CONTROL CENTRE

NOTE 1
1/700A
4000/ 5A
5VA
5P20
00F-01
4000/ 5A
30VA
C10.5
00F-08
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
AFD
ACB
2A
4000A
50
51
50G
51G
24VDC CONTROL POWER SUPPLY
NOTE 7
125A
160A
SD
NOTE 6

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 15/09/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

415V 3NE~ 50Hz 4000A 65kA for 1sec

4F-01	4F-11	5R-17	3F-01	3F-05	3F-09 INC	1R-05	3F-13	1F-01	1F-05	1F-09	1F-13	
100A	100A	20A	32A	32A	32A	630A	32A	32A	32A	32A	32A	
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	
/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >
100/1A	90A	70A	70A	70A	70A	630A	70A	70A	70A	70A	70A	
		EOL	EOL	EOL	EOL		EOL	EOL	EOL	EOL	EOL	

FEEDER TO F13-SW-102
2000-F13-SCM-EL-10324

E-SHUT
F13SR001-LCS
NOTE 12

CONTINUED ON SHT 2
2000-F13-SLD-EL-10302

M 3~	CP	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	
37kW	29KW	0.37kW	7.5kW	7.5kW	7.5kW	7.5kW	7.5kW	7.5kW	7.5kW	7.5kW	7.5kW	
F13-CV-404	F13CU407-CP	F13-FN-425	F13-FN-427	F13-FN-428	F13-FN-429	F13-FN-430	F13-FN-431	F13-FN-432	F13-FN-433	F13-FN-434		
RECYCLE CONVEYOR 1	SAG MILL VSD COOLING SYSTEM	SAG MILL MOTOR LUBE SKID COOLER FAN	SAG MILL MOTOR 1 FAN 1	SAG MILL MOTOR 1 FAN 2	SAG MILL MOTOR 1 FAN 3	SAG MILL MOTOR 1 FAN 4	SAG MILL MOTOR 2 FAN 1	SAG MILL MOTOR 2 FAN 2	SAG MILL MOTOR 2 FAN 3	SAG MILL MOTOR 2 FAN 4		
2000-F13-SCM-EL-10311	2000-F13-SCM-EL-10310	2000-F13-SCM-EL-10312	2000-F13-SCM-EL-10313	2000-F13-SCM-EL-10314	2000-F13-SCM-EL-10315	2000-F13-SCM-EL-10316	2000-F13-SCM-EL-10317	2000-F13-SCM-EL-10318	2000-F13-SCM-EL-10319	2000-F13-SCM-EL-10320		

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F13-SW-002 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F13-BLK-IC-10002.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- / > — EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA ST TRIP SET POINT / xxxA CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- 1 E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F13-SLD-EL-10301_Rev1

SAG mill F13-SW-002 415V MCC single line diagram sheet 1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCOM GROWTH PROJECT\03 DELIVERABLES\ELV\F13 - MILLING & CLASSIFICATION\2000-F13-SLD-EL-10301DWG 27AUG25 7:50AM HOTSEAT04